

● IQOO CITY BATTLES 2026 · HYDERABAD · SMART LIVING

Taar.

Hold a phone near a cable. It tells you whether current is flowing — **no probe, no contact, no network.**

Team **Against Default** · Goutham Pakala, Geetha Tigulla

● THE PROBLEM

The faults that start fires are invisible until they aren't.

A loose connection arcing behind a cover plate. A circuit run permanently over its rating. Neither shows itself until something burns.

The people who would catch them — small electricians, facility technicians — are diagnosing by feel. The instruments that would make it visible are a clamp meter, a thermal camera and an arc-fault analyser.

That is a serious capital outlay for a two-person firm. So the tools stay with large contractors, and the buildings that most need checking never get checked.

And a safety problem underneath

Before working on a circuit, an electrician needs to know it is genuinely dead. The common method is to probe it — or to trust a label on a board that may be a decade out of date.

Every phone already has the instruments

A magnetometer that senses current. A microphone that hears an arc. Nobody has pointed them at this.

Two physical facts, one three-second capture.

Magnetometer → current

$$B = \mu_0 I / 2\pi r$$

A current-carrying conductor broadcasts its current as a magnetic field. At 3 cm, 5 A produces about 33 μT — against an Earth background of 45 μT and a sensor floor we measured at 0.4 μT .

The principle a clamp meter uses, without the clamp.

Microphone → fault

$$100 \text{ Hz} = 2 \times \text{mains}$$

A series arc re-ignites twice per mains cycle, so its broadband noise is amplitude-modulated at 100 Hz. We look for energy at that frequency, not for energy in general.

Against four confounders, a plain energy detector fired on every one. This did not.

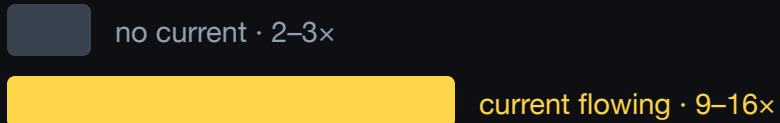
Both run on the device. Boards live in basements with no signal; a building's circuit layout is not a contractor's data to upload; and a measurement of response time cannot have a network round trip inside it.

● MEASURED, NOT ASSUMED

We built it, put it on a phone, and tested the claim.

SIX CAPTURES · FRIDGE SUPPLY CABLE · PHONE TAPED IN PLACE

STATE	CONTRAST AT 50 HZ
No current	3x · 2x · 2x
Compressor running	9x · 16x



A factor of three between the populations, no overlap. The detection threshold is now set from that separation, replacing one chosen from simulation that sat close enough to the idle readings to call a dead circuit live.

105.3 Hz

MAGNETOMETER, 21 DISTINCT PHASES

0.40 μ T

NOISE FLOOR, REPEATABLE ACROSS SESSIONS

Honest scale: six captures, one phone, one appliance, ~1–1.5 A through a twin cable whose conductors largely cancel. Enough to establish the premise. Not enough to quote a sensitivity figure — so we don't.

Four defects that simulation never found.

01 Two Android clocks

`SensorEvent.timestamp` includes deep sleep; `System.nanoTime()` does not. Comparing them made every capture return nothing at all.

02 Vector magnitude destroyed the signal

Reducing three axes to $\sqrt{x^2 + y^2 + z^2}$ hid the AC component behind Earth's static 45 μT — a 1 μT field moved the magnitude by 0.01 μT . Each axis is now fitted separately.

03 Detection normalised by total variance

A hand-held capture drifts several μT . That drift buried the 50 Hz term and reported a real signal as zero confidence. Now measured against neighbouring frequencies instead.

04 Thresholds guessed rather than measured

The arc statistic's floor was five times too small, and reported a domestic fridge as an arcing fault.

Each is now covered by a regression test. A simulation that had predicted any of them would have been a better simulation. It didn't — which is the argument for building early and putting it on real hardware.

• ALREADY BUILT

Not a concept. A working Android app.

44

OF 44 LOGIC CHECKS PASS

9.1 MB

DEBUG APK, BUILDS CLEAN

0

NETWORK PERMISSIONS

Signal path

Per-axis sine fit at 50 Hz, detrended, amplitude as the vector sum →
Lomb–Scargle contrast against 30–48 and 52–70 Hz → scoring by
median and MAD against the circuit's own reference.

On top of it

A rules table over six faults, each with its evidence and an action in
English and Telugu. A nearest-centroid classifier trained on the
technician's own labels. Qwen3-1.7B on the Hexagon NPU for patterns
the rules don't recognise.

What Taar does not do.

It is a triage aid, not a certification

It does not replace a licensed electrician, a calibrated clamp meter, an insulation tester or a statutory inspection. The app says so on the result screen, not in a settings page.

The acoustic half is designed, not validated

No real arc has been recorded. The magnetometer half rests on textbook physics and now on measurement; the arc half does not yet.

It reports an index, not amperes

Unless the circuit has been calibrated against a known load. A fabricated ampere figure is worse than none.

It refuses rather than guesses

No reference, or an ill-conditioned fit, produces a refusal. An electrician acting on a confident wrong reading is the worst outcome this project has.

Our pre-event work is a research spike on a separate branch, rebuilt from an empty scaffold during the event window and never merged. The protocol is written down, and we will confirm it with an organiser at check-in.

• THE DEMO

Ninety seconds. Airplane mode on, visibly.

Appliance off

Phone taped to the cable. **Measure.** No mains detected.

Appliance on

Nothing moves but the switch. **Measure.** Contrast jumps, load index climbs.

The point

A phone detects electricity through insulation, touching nothing.

Yesterday that was a claim. Today there are **six captures** behind it.